Basis of image analysis for biology
Part 1

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9-10;30 am

MScBOOST

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-Main parameters defining digital images

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Qualitative and quantitative analysis

Moment of action: analyze a movie

-Look for metadata information: time scale

-Image preparation

-Image visualization

-Image segmentation based on pixel intesnity

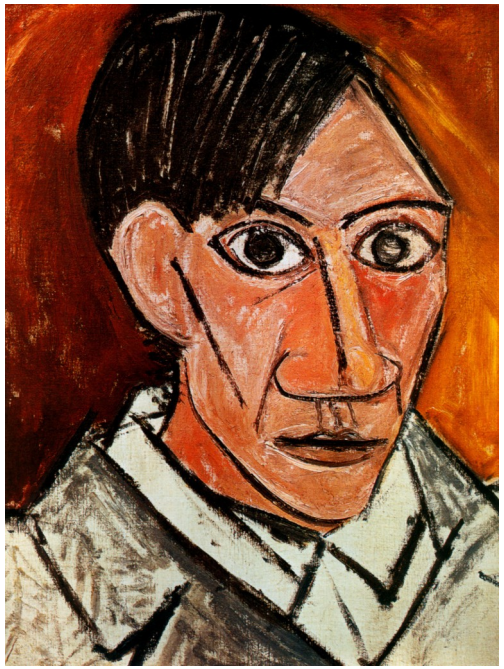
1. Introduction to image analysis

What is an image?

An image is a representation of an object (person, nature, inanimate objects…)

Picture = digital image = image stored in binary form and made of pixels

Picasso auto portrait



Picasso digital selfie

1. Introduction to image analysis

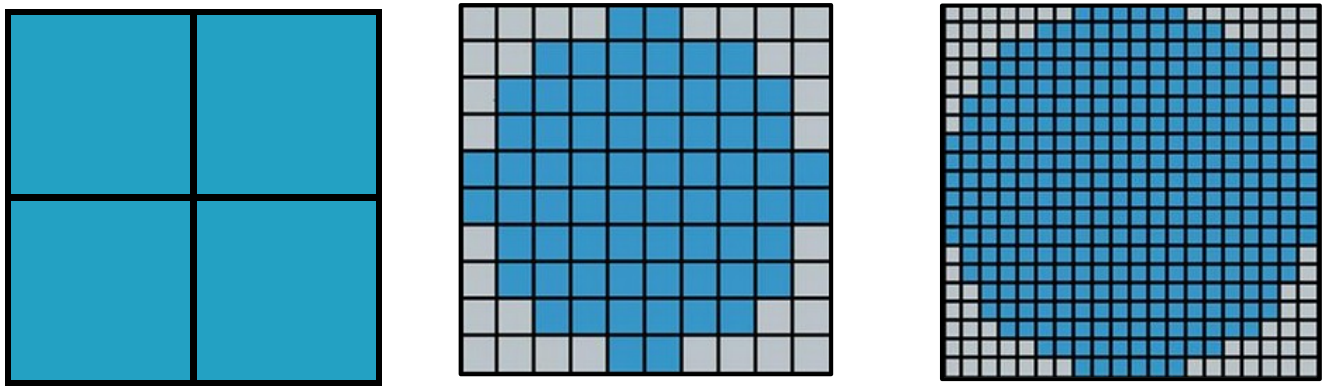
Parameters defining a digital image: pixel (px)

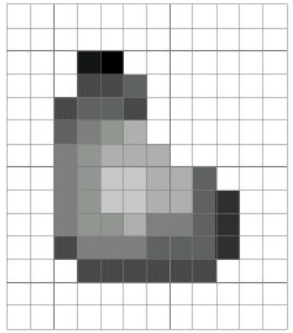
A pixel is the smallest unit of digital image. Pixels are combined to form a complete image, video, text, or any visible thing on a display

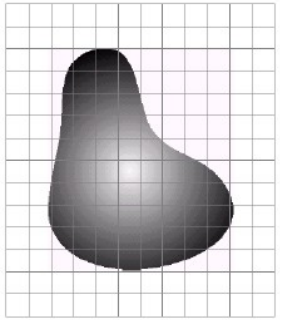
Pixel dimension

Image size
1 inch
(2,54 cm)

	2x2 px	10x10 px	20x20 px
Resolution:	2 px/ inch	10 px/ inch	20 px/ inch



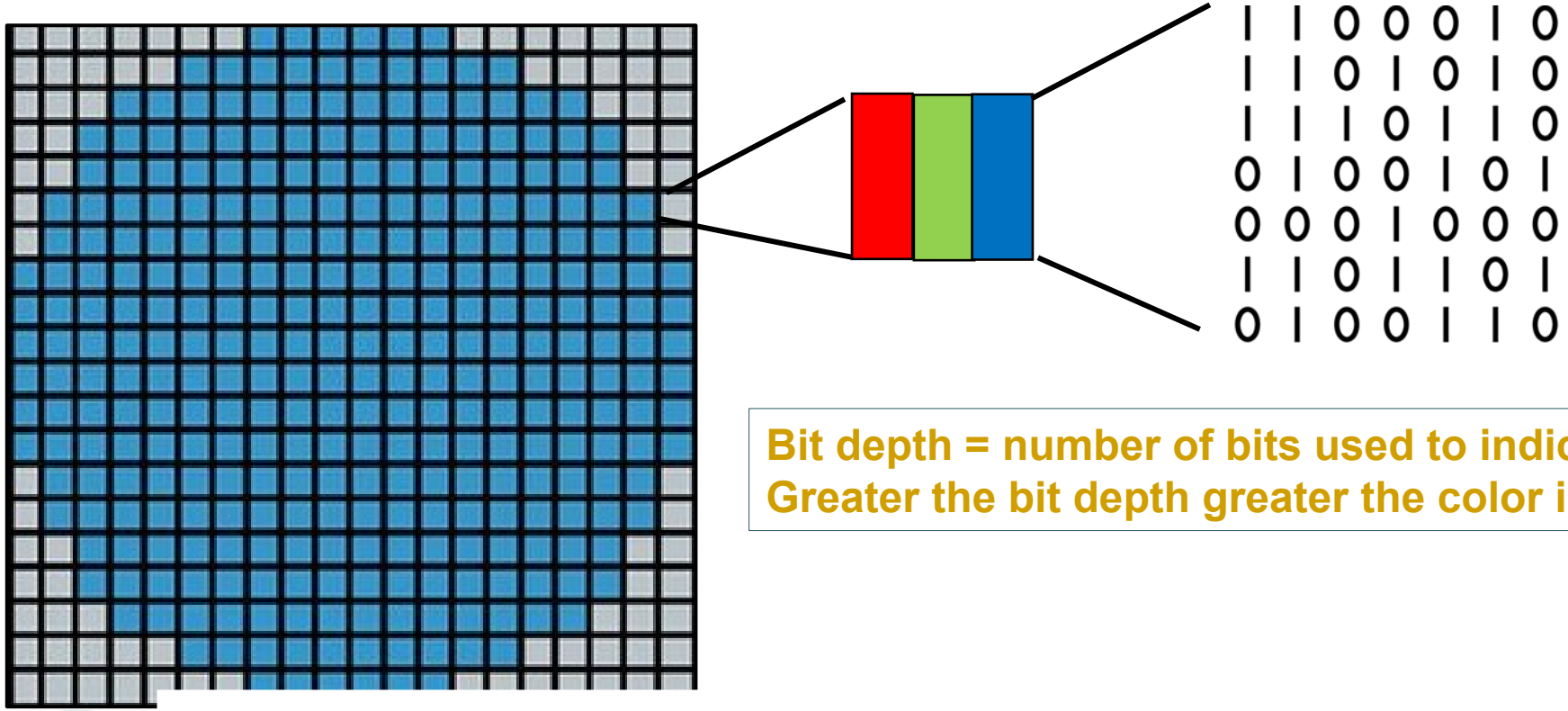




<u>The resolution increases with the number of pixel/inch</u>

Parameters defining a digital image: bit or color depth

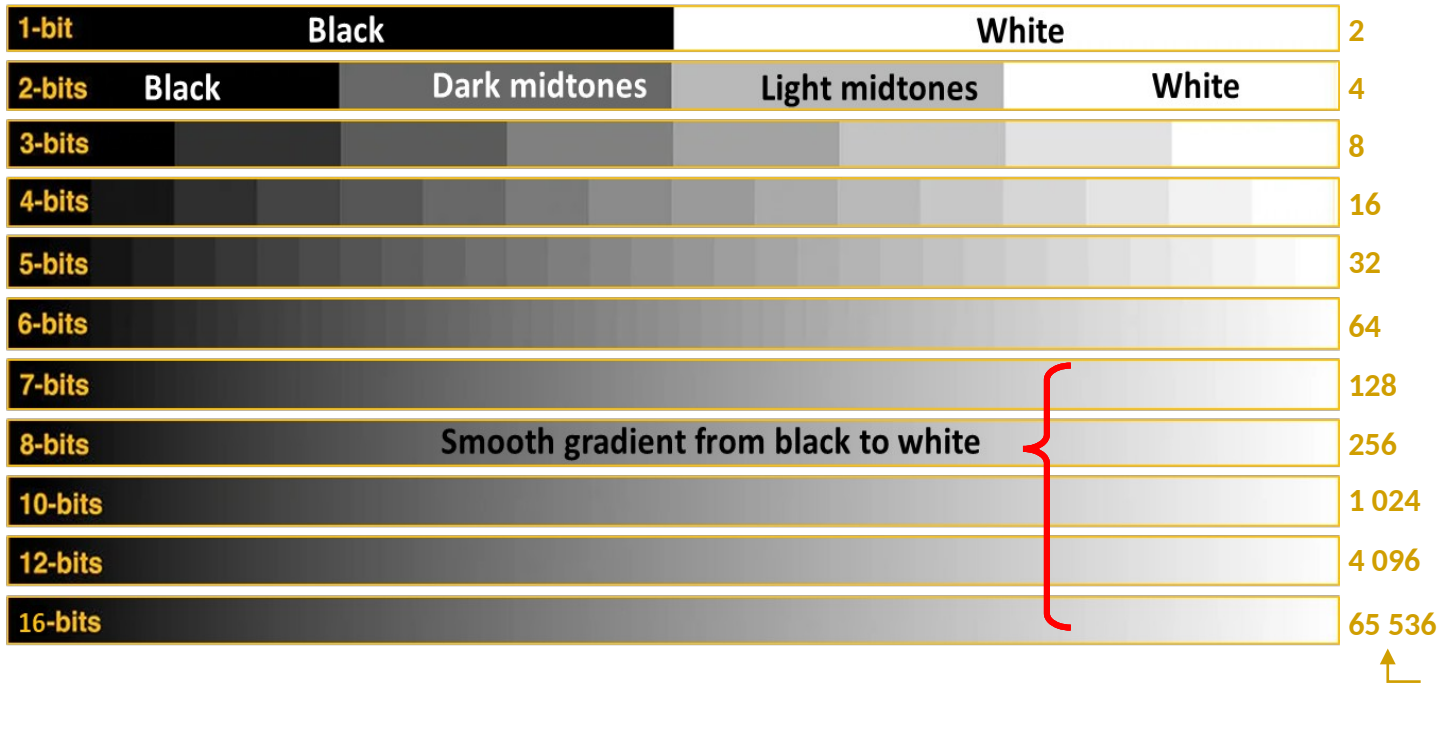
The color of each pixel is determined by the specific blending of **RED, GREEN and BLUE** (RGB) subpixels.
Information of color is stored in form of binary code (1,0); the quantity of binary units or **bit** used to describe each pixel is called **bit depth**



Bit depth = number of bits used to indicate the color of a single pixel.
Greater the bit depth greater the color information of a pixel

1. Introduction to image analysis

Parameters defining a digital image: bit or color depth



For a grayscale image, the **bit depth quantifies how many unique shades are available**. Images with higher bit depths can encode more shades or colors since there are more combinations of 0's and 1's available

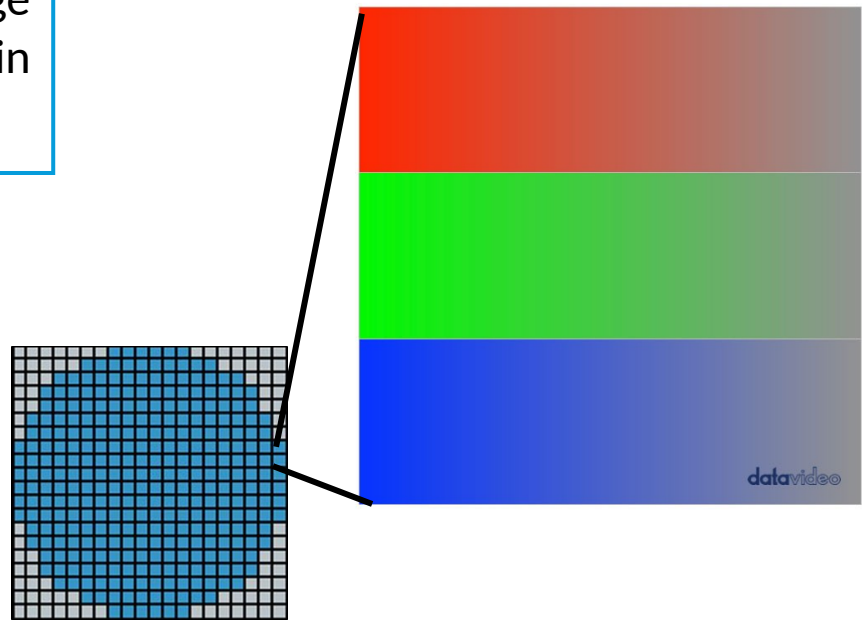
http://www.comp.dit.ie/bmacnamee -/materials/dip/lectures/ImageProcessing1-Introduction.ppt
https://petapixel.com/2018/09/19/8-12-14-vs-16-bit-depth-what-do-you-really-need/

1. Introduction to image analysis

Parameters defining a digital image: bit or color depth

8 bit depth: High precision image (256 shades of black and white in greyscale)

In color scale millions of color combination !



256
x
256
x
256

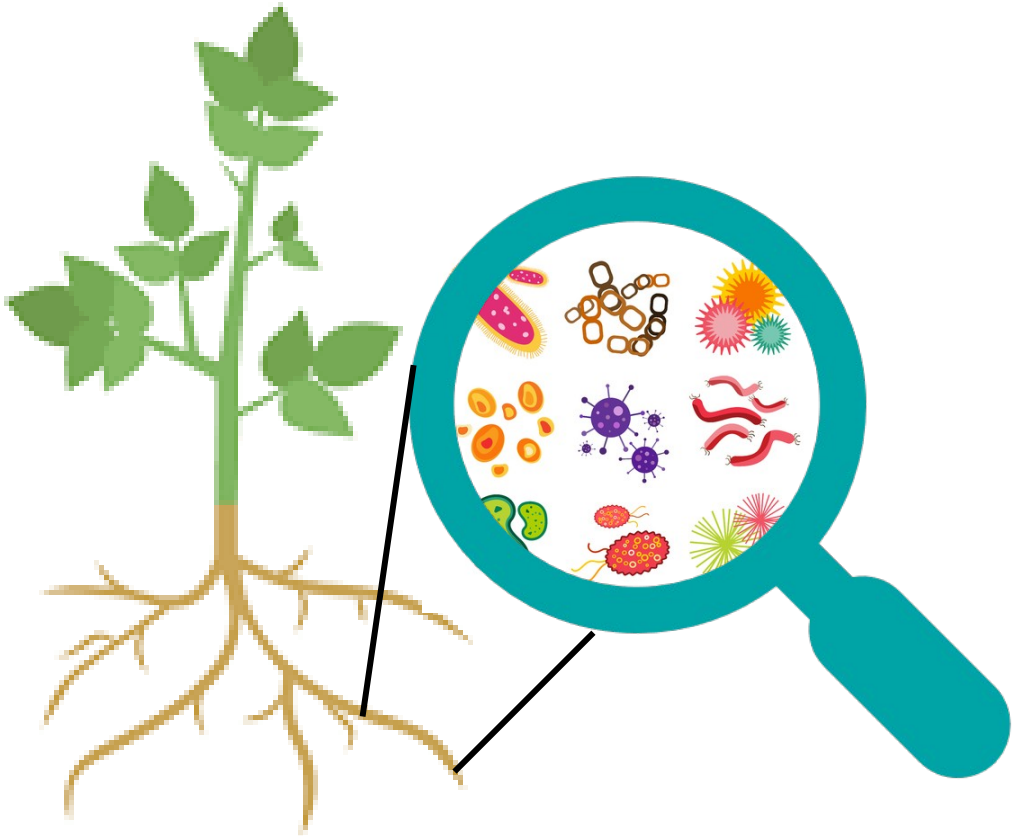
→ 16,777,216

<u>8 bit depth images are required for image analysis in many sfotwares</u>

1. Introduction to image analysis

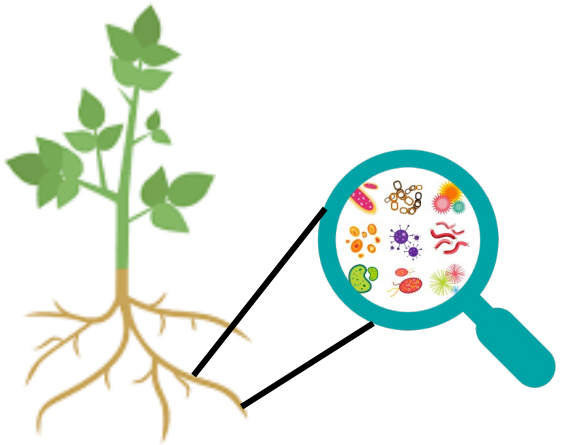
How to acquire a digital image?

...Different for microscopic world



1. Introduction to image analysis

How to acquire a digital image for microscopic objects?



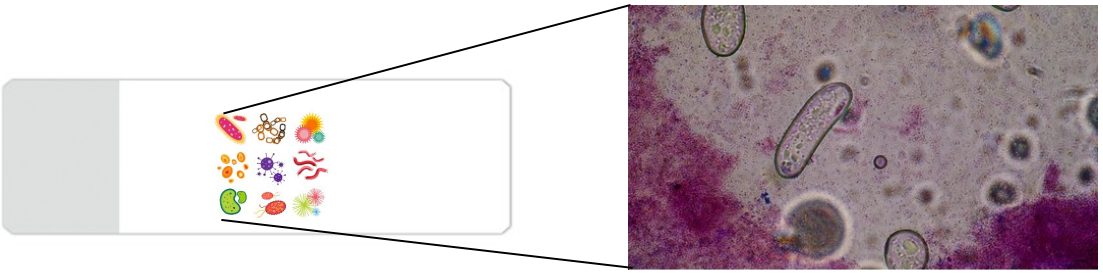
1. Prepare your sample

Epifluorescence microscope

2. Place it in the microscope plate and observe it

3. Take images with microscope-adapted cameras

4. Save the images in suitable analysis formats (tiff, jpeg...)

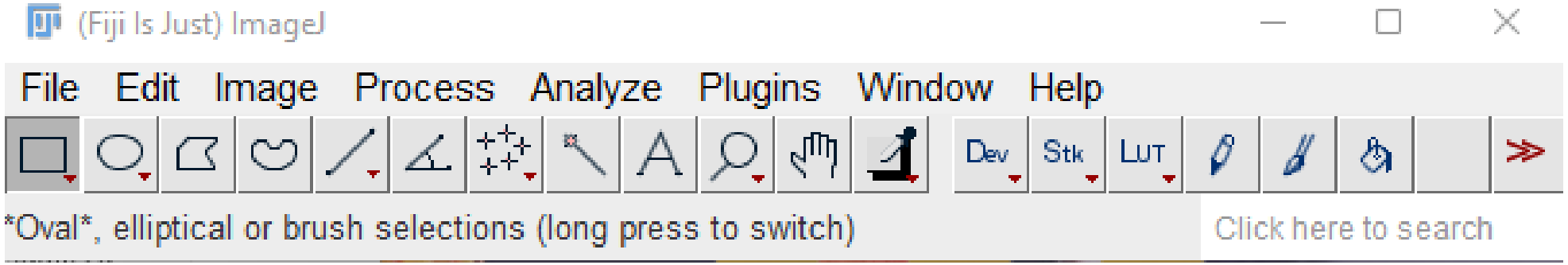


1. Introduction to image analysis

How to analyse the digital scientific image obtained?

- Fiji software

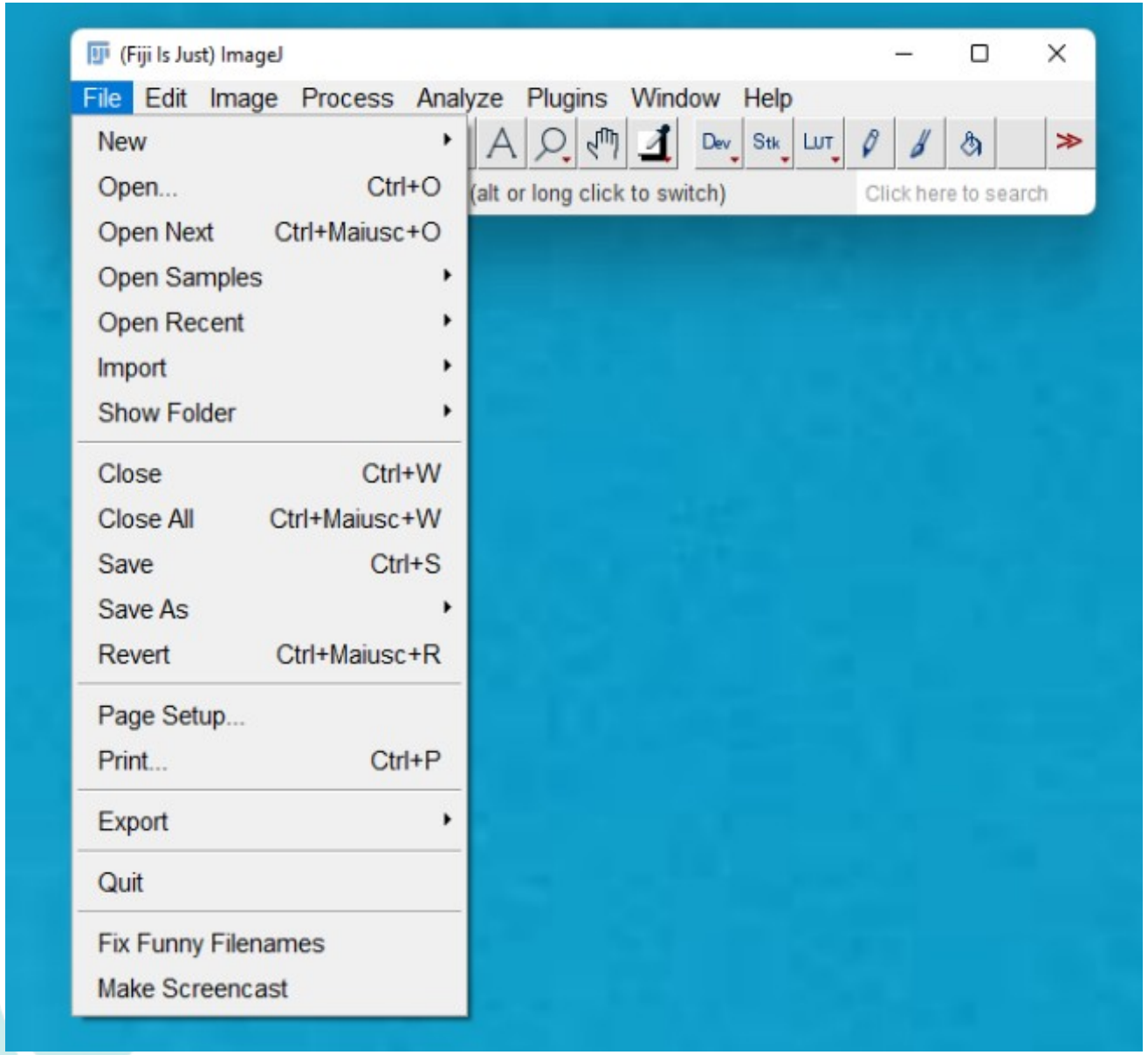
Fiji is an image processing package bundling many plugins which facilitate scientific image analysis (user friendly)



… A “photoshop” for scientific image analysis ;)

1. Introduction to image analysis

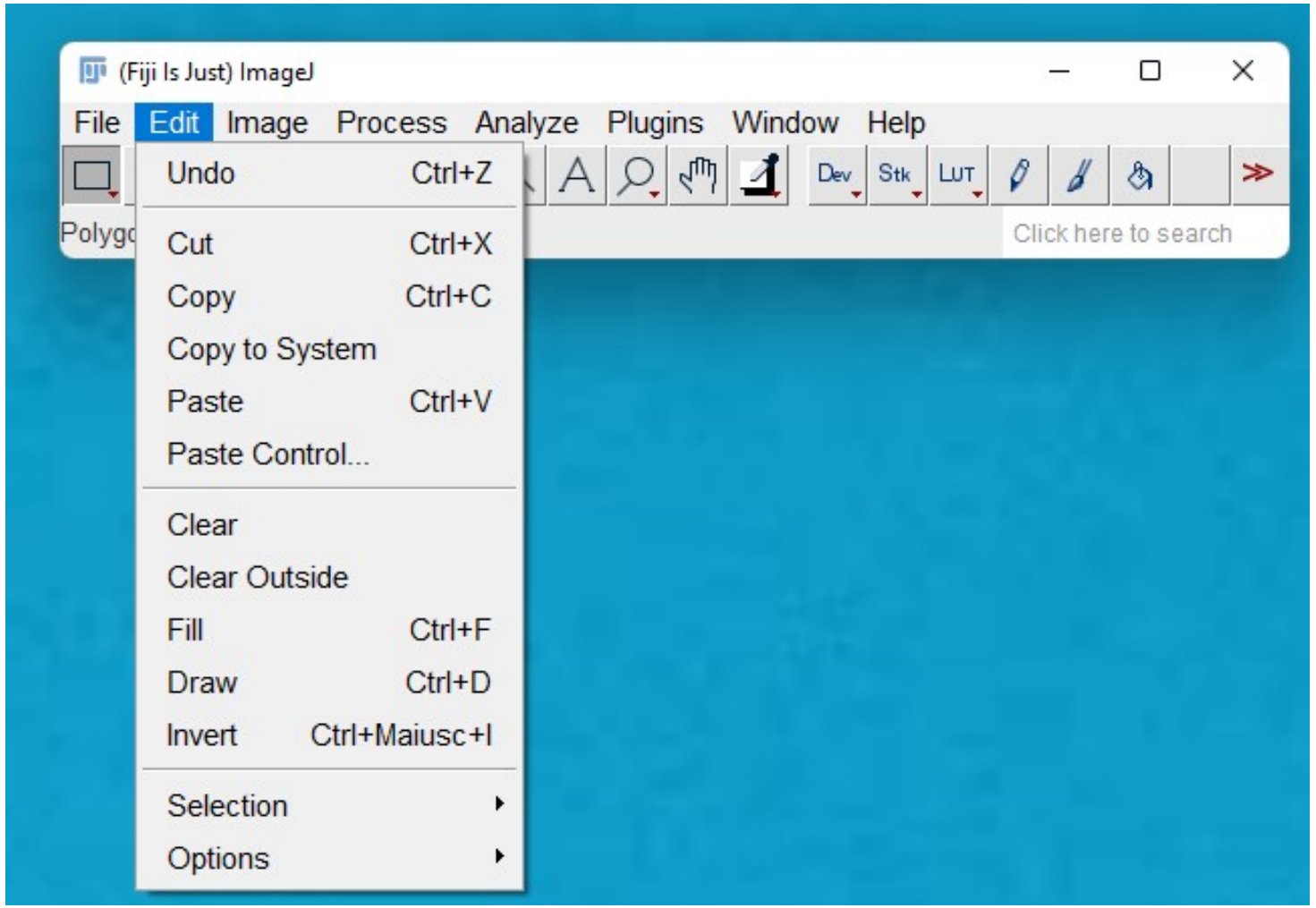
- File Menu



Open files; import image sequences; quick access to recent opened files; save images in different formats…

1. Introduction to image analysis

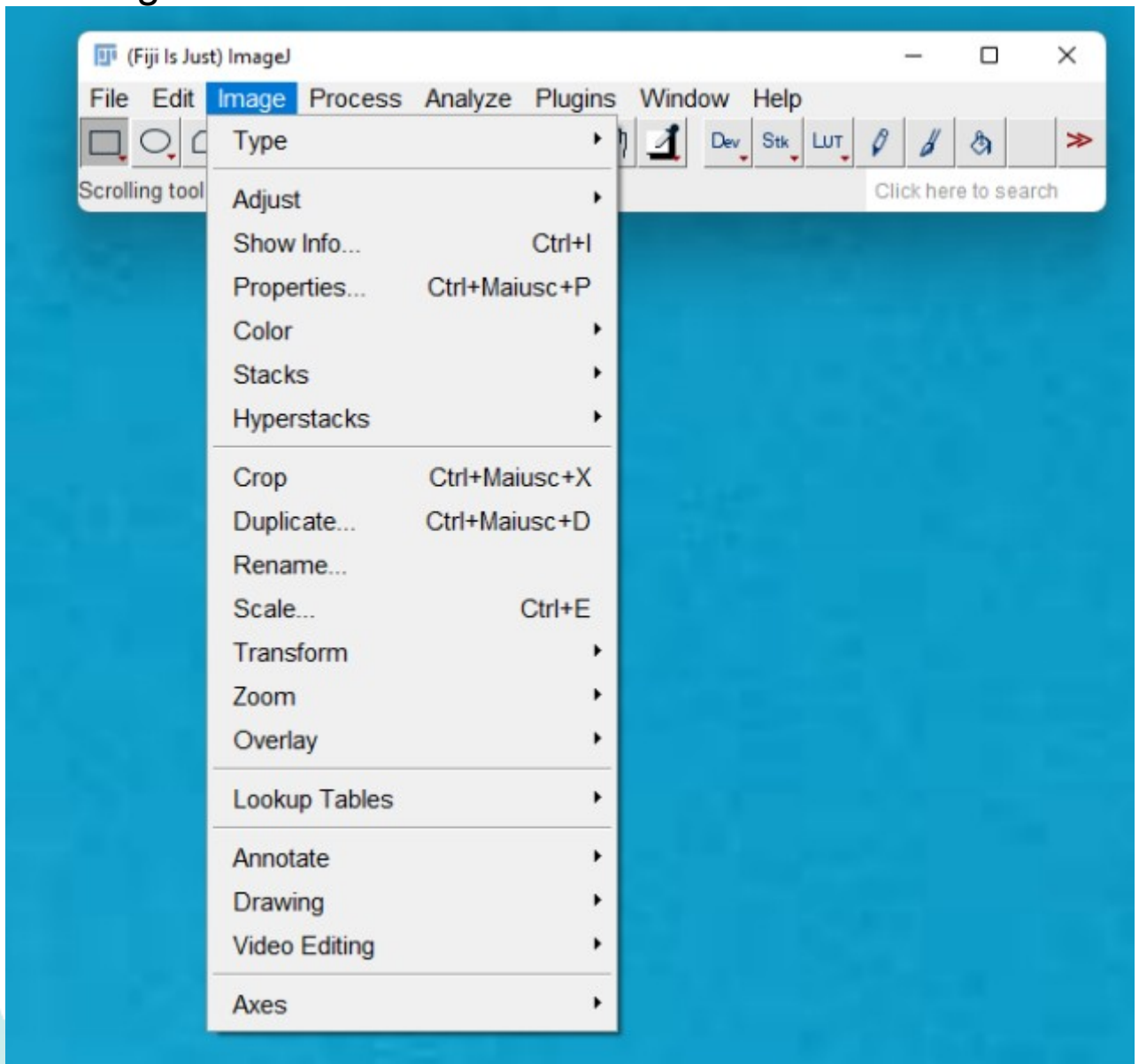
- Edit Menu



Copy and paste objects; cut areas; draw objects; undo actions…

1. Introduction to image analysis

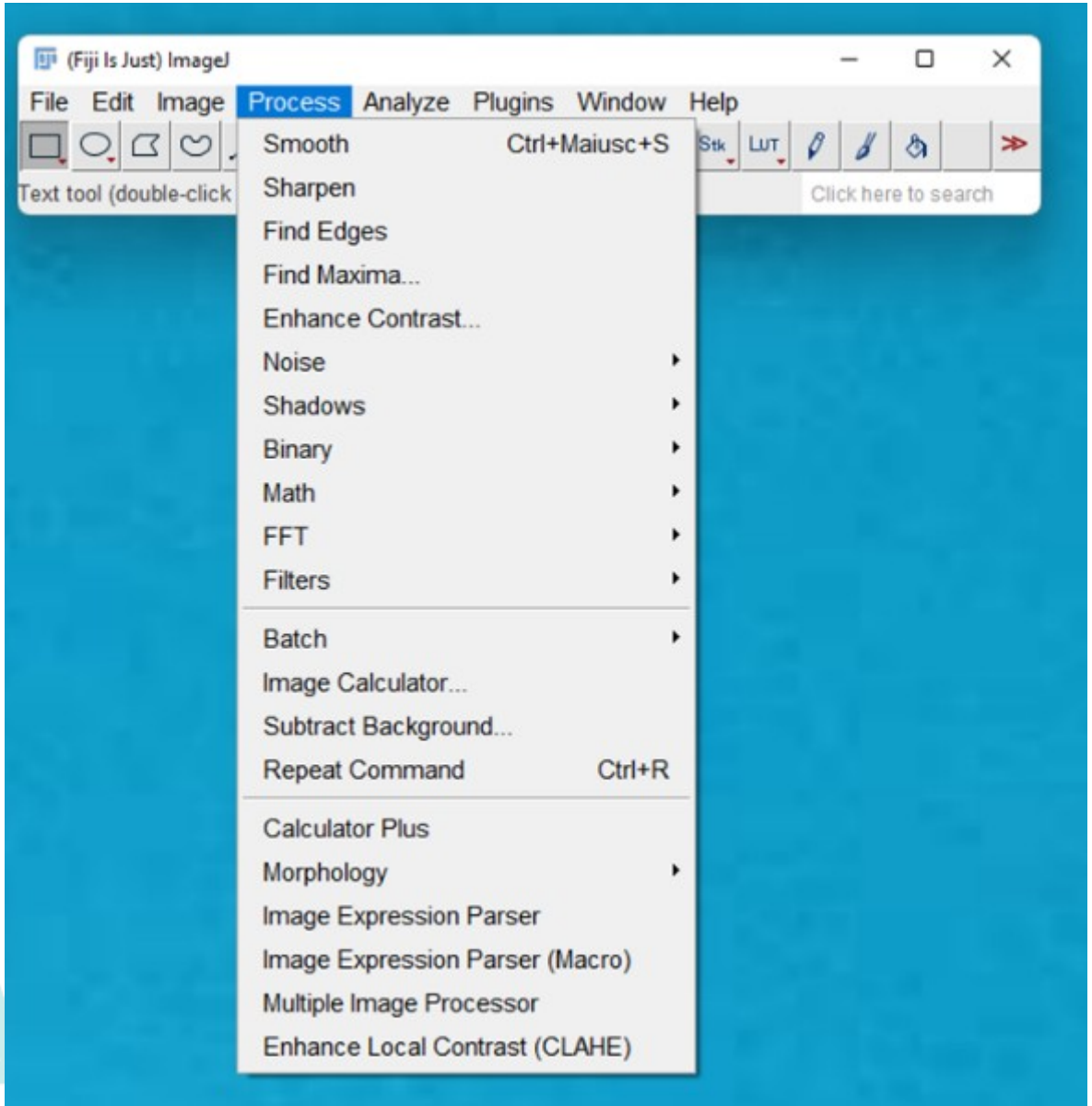
- Image Menu



Type (8-bit,16-bit…); adjust parameters (brightness/contrast, threshold); image Info (metadata); properties (channels, scale, frames…); create stacks with different images; crop selected areas; duplicate images…

1. Introduction to image analysis

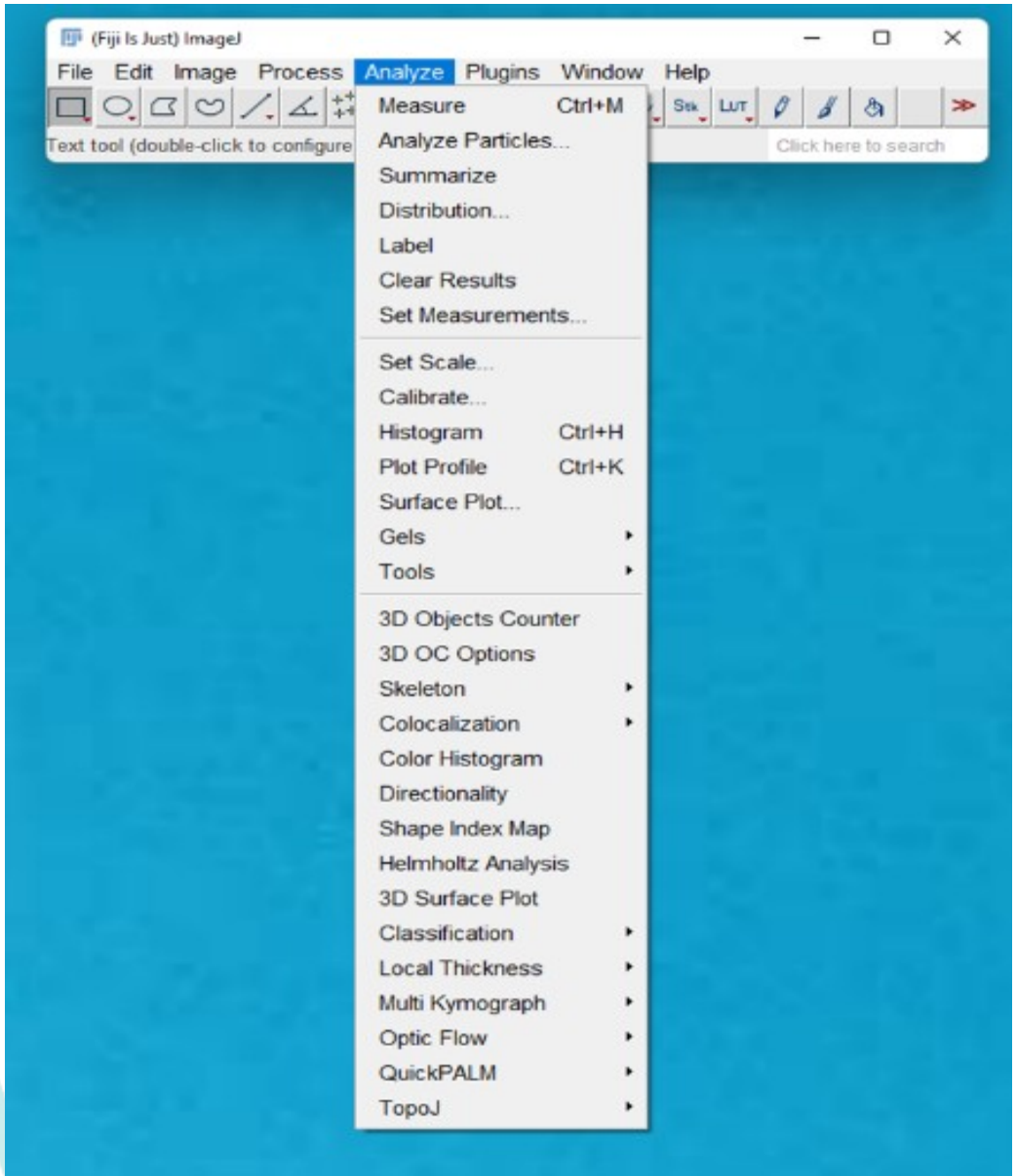
- Process Menu



Different commands to perform image segmentation and morphological enhancement by sharpen objects, finding edges, enhancing contrast, making the images binary, subtracting objects from background…

1. Introduction to image analysis

- Analyze Menu

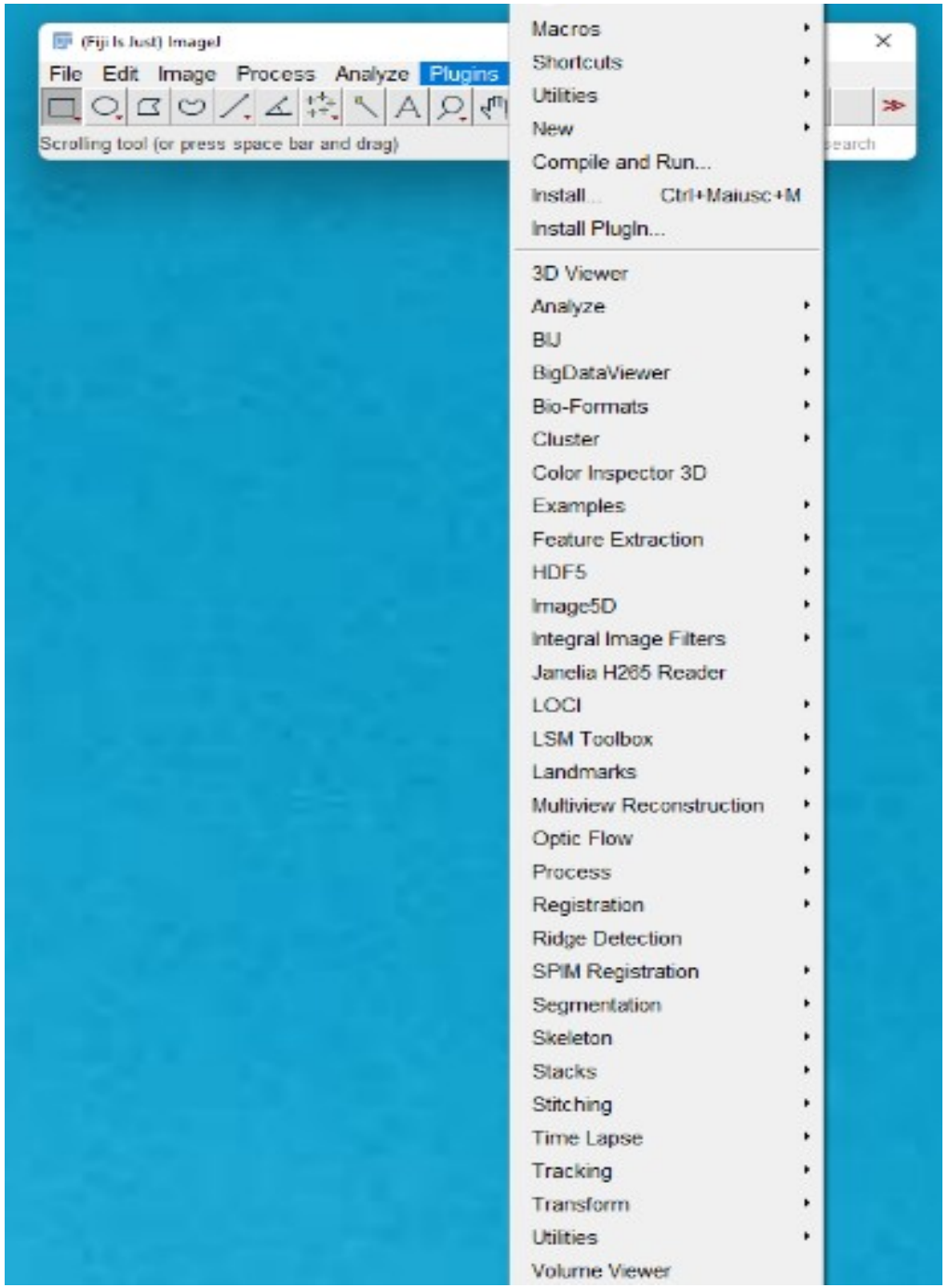


Set scale and measurement for the analysis of the image; calculate histogram of pixel distribution (in case of non-binary images); create grids; dimensional analysis (kymograph)…

1. Introduction to image analysis

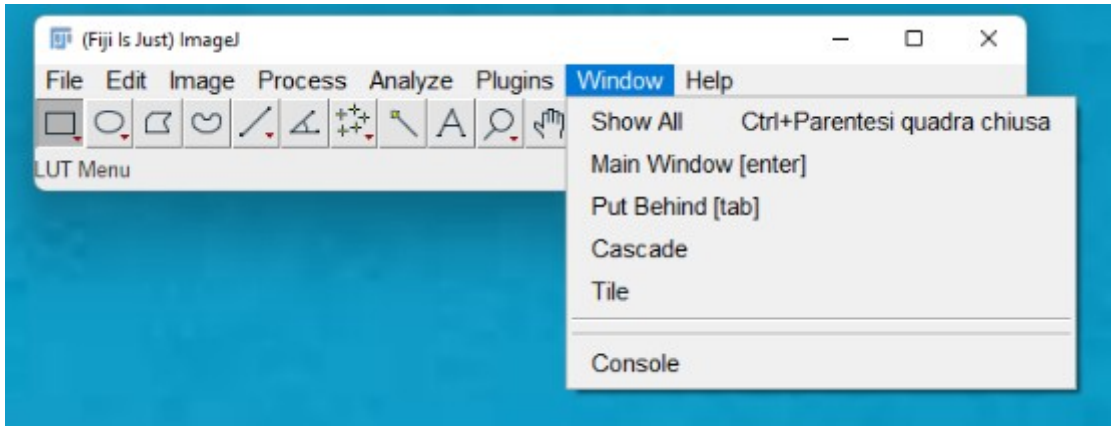
- Plugins Menu

Available loadable code modules for acquisition, analysis, segmentation, graphics, programming, tracking ….
Possibility of installation of new plugins
(https://imagej.nih.gov/ij/docs/menus/plugins.html)

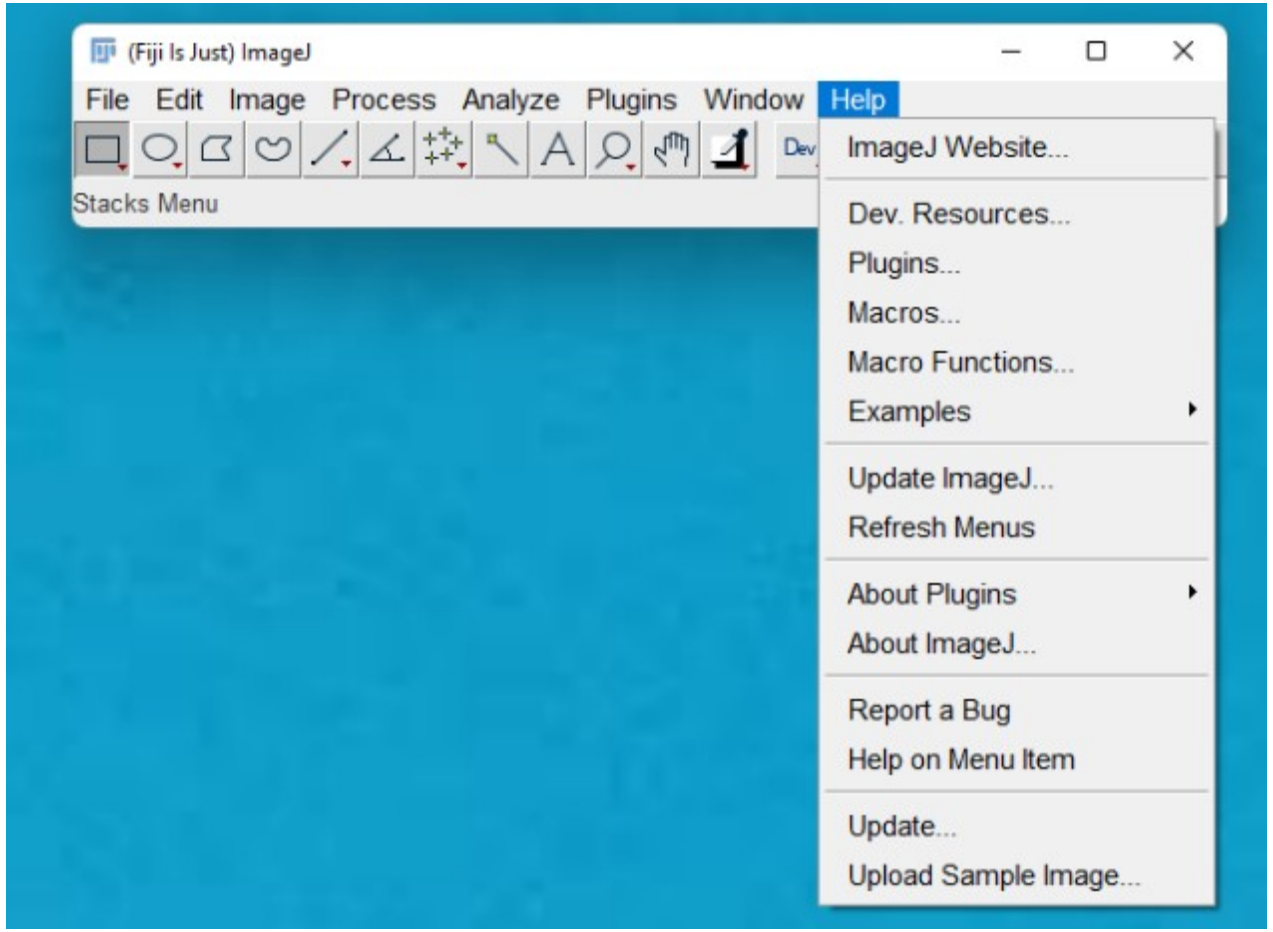


1. Introduction to image analysis

- Window and Help Menu



Show windows open and prioritize some windows



Access to ImageJ website; access to available plugins and to download; updates…

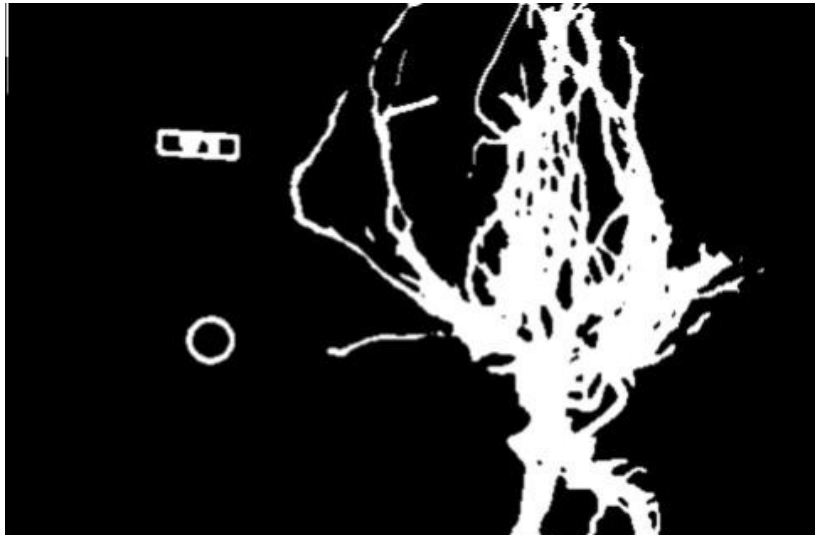
2. Bases of image analysis

- Which analysis can we do?

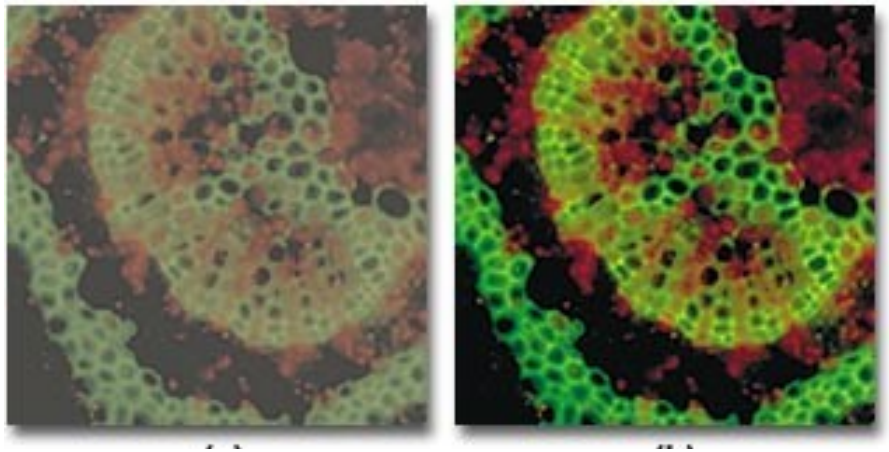
Qualitative

Binary threshold Binary Inverse Technique

Plant root



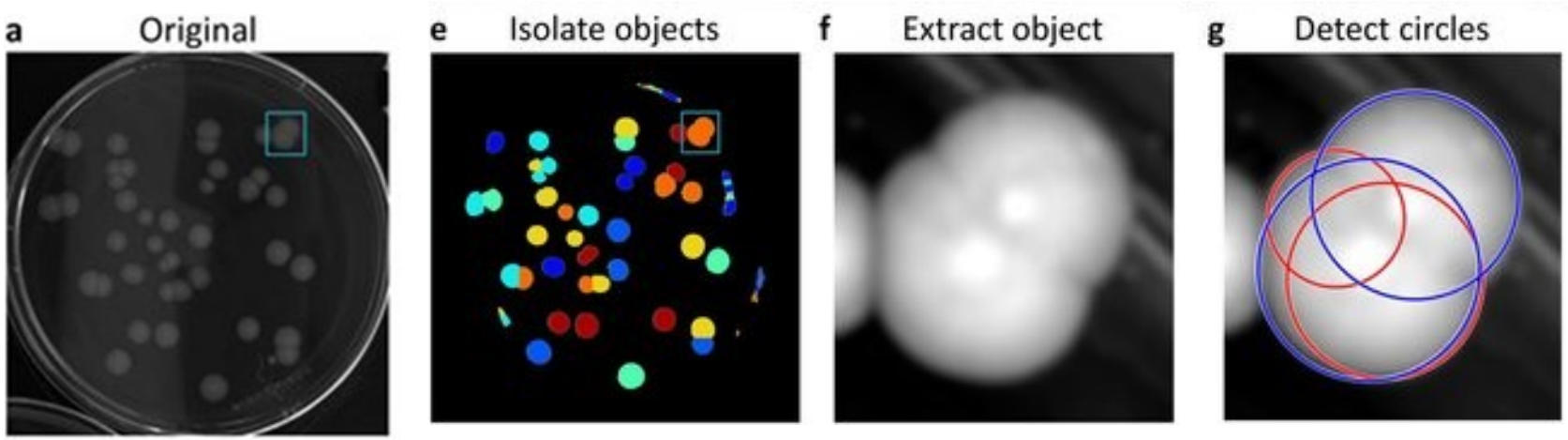
Plant Vascular bundle



2. Bases of image analysis

- Which analysis can we do?

Quantitative



DOI: 10.1038/s41598-020-72979-4

Automatically count fused cells

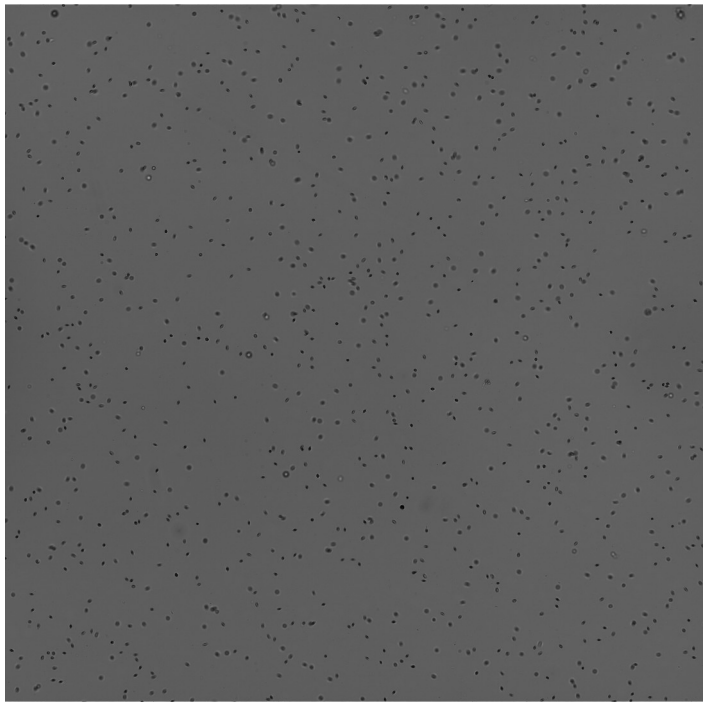
2. Bases of image analysis

Moment of action : lets make some image analysis

We are going to analyze “Movie_training” (set of digital images)

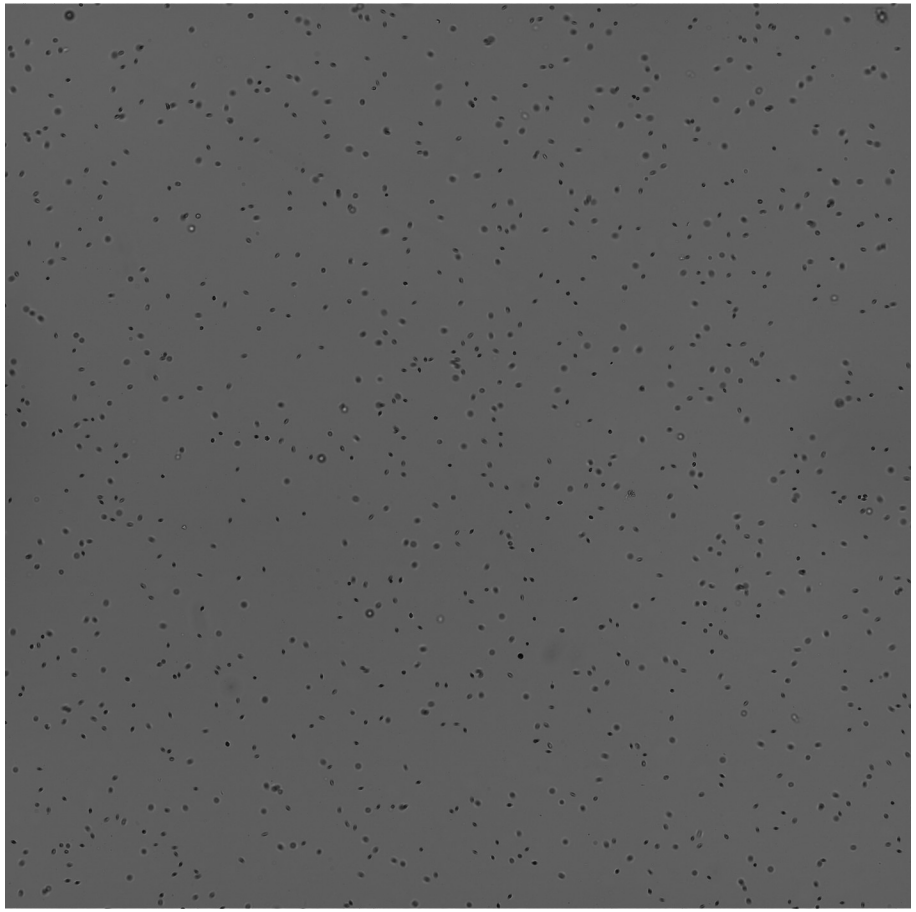
Microscope Axio-observer Z1
Lens X5

Movie of swimming cells (zoospores)
Zoospore diameter = 10microns



16 bit

2. Bases of image analysis



Our objective today is to enhance the visualization of the cells and count them

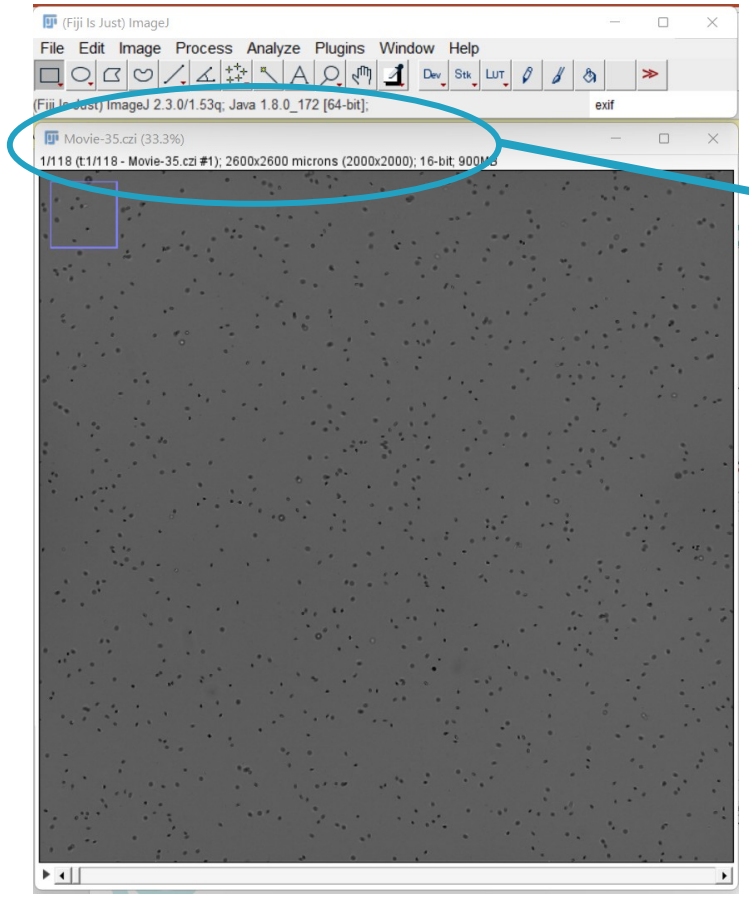
2.1 Required information from metadata

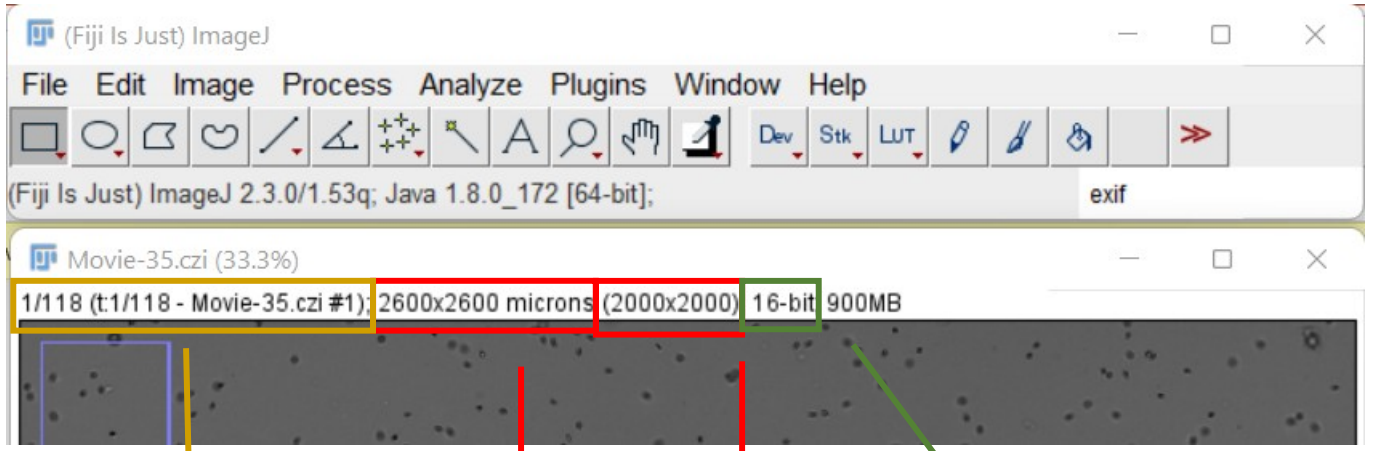
2.2 Image processing with Fiji

2.2.1 Image preparation
2.2.2 Image better visualization
2.2.3 Image segmentation

2.1 Required information from metadata

Where can you find these information? Fiji metadata





Number of frames

NOT the time scale

Image size microns

number of pixels

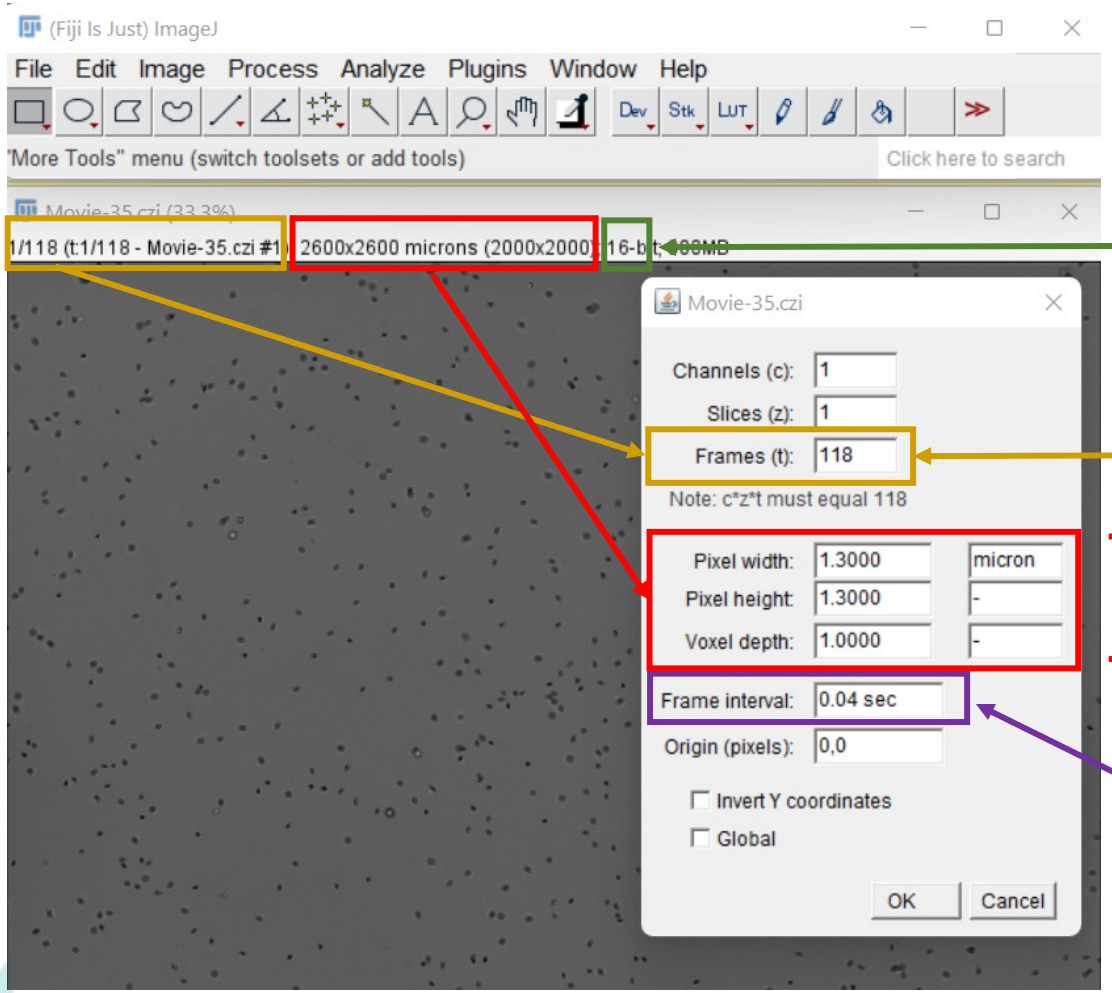
Calculate the resolution
1 pixel = 1,3 microns

Depth of bit
Per pixel

2.1 Required information from metadata

Where can you find these information? Fiji metadata

Fiji : Image > Properties



Bit depth per pixel

Number of frames

Pixel size
1 pixel = 1,3 microns

Time scale : 40ms
(Needs validation in the acquisition software)

Keep these Information for later.

Bit depth
Number of frame
Pixel size
Time sclale

2.2 Image processing with Fiji

2.2.1 Image preparation

Convert to 8 bit

↓

Better visualization

Image Enhancement / sharpening

↓

Image treatment

Segmentation and Morphological Processing

↓

Representation & Description

8-bit
Bit depth required for Fiji

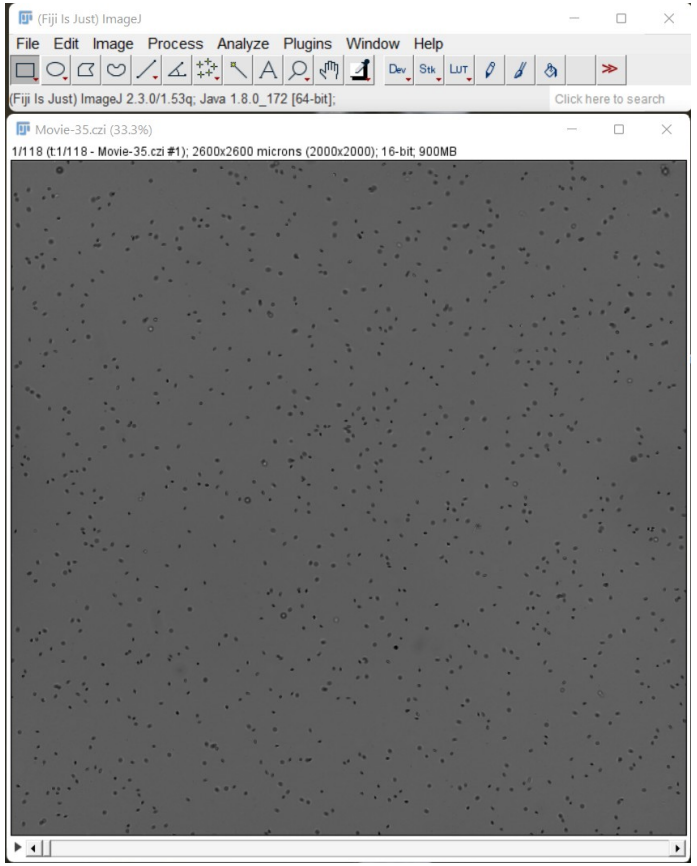
Adapted from : https://benchpartner.com/describe-the-block-diagram-of-digital-image-processing-system https://buzztech.in/fundamental-steps-in-digital-image-processing/

2.2 Image processing with Fiji

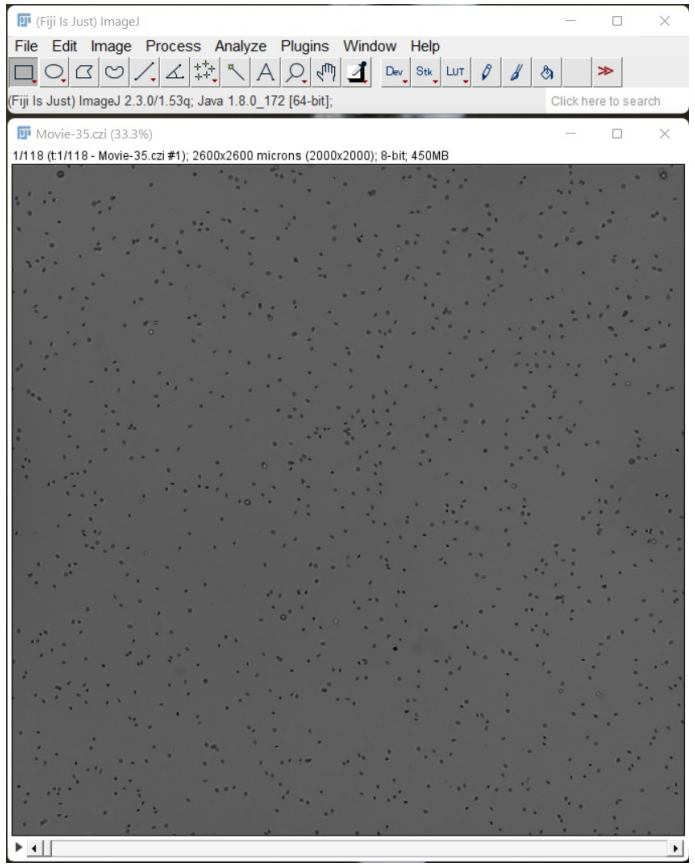
2.2.1 Image preparation

Image>type>8-bit

8-bit Is required for Fiji



16-bit



8-bit

Adapted from : https://benchpartner.com/describe-the-block-diagram-of-digital-image-processing-system https://buzztech.in/fundamental-steps-in-digital-image-processing/

2.2 Image processing with Fiji

2.2.2 Better image visualization

Convert to 8 bit

↓

Better visualization

Image enhancement / sharpening

↓

Image treatment

Morphological processing and segmentation

↓

Representation & Description

Create a better image
and
Observe objects weakly visible

↓

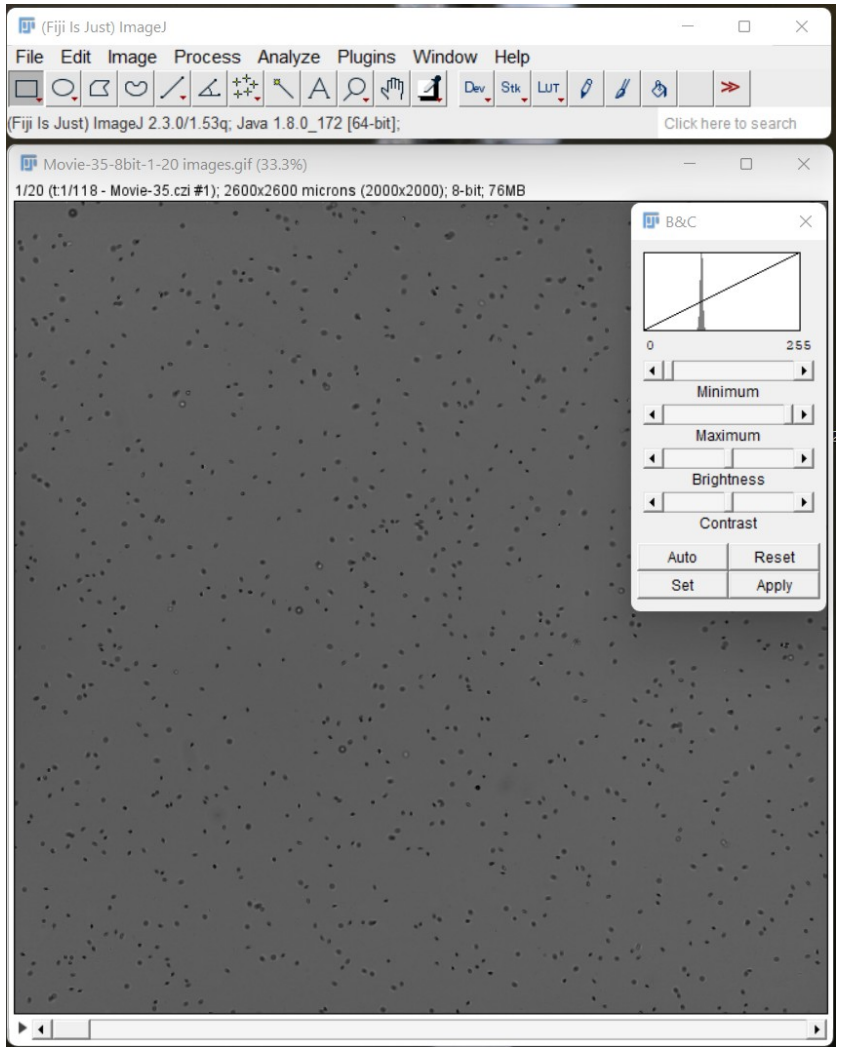
Adjust Brightness,
Noise removal,
Background treatment,
…

Adapted from : https://benchpartner.com/describe-the-block-diagram-of-digital-image-processing-system https://buzztech.in/fundamental-steps-in-digital-image-processing/

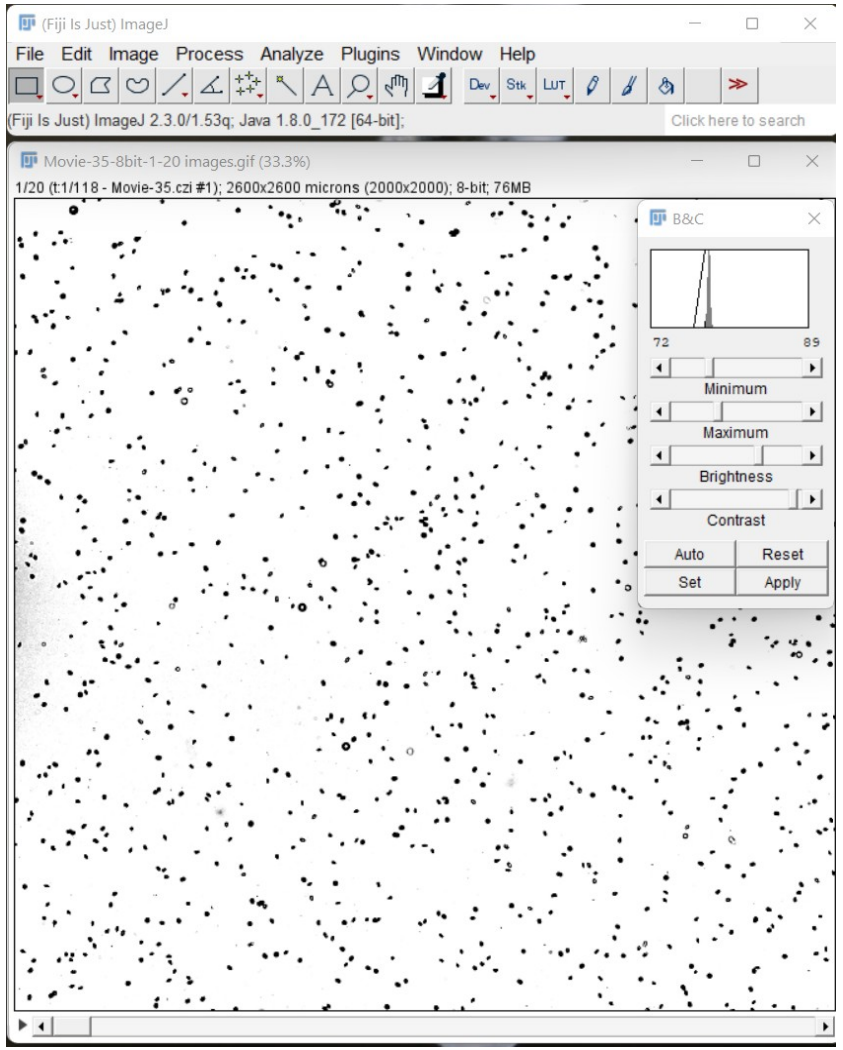
2.2 Image processing with Fiji

2.2.2 Better Image visualization:

Fiji : Image > Adjust brightness/contrast



Set parameters for better visualization



2.2 Image processing with Fiji

Convert to 8 bit

Better visualization

Image Enhancement / sharpening

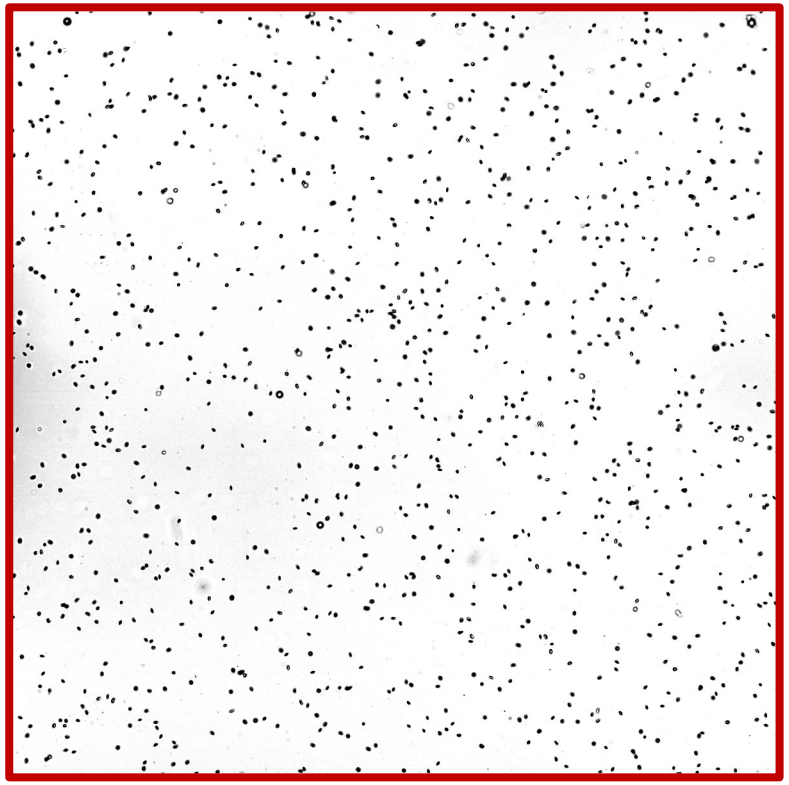
Image treatment

Segmentation and Morphological Processing

Representation & Description

Movie of swimming cells (zoospores)

Microscope Z1; Conventional camera; 5x



→ **Respond to our scientific questions**

-How many cells are there?

2.2 Image processing with Fiji

2.2.3 Image treatment : segmentation

Convert to 8 bit

↓

Better visualization

Image Enhancement / sharpening

↓

Image treatment

Segmentation and Morphological Processing

↓

Representation & Description

Extract image components that are useful in the analysis.

Extract the objects (or particles) based on their properties (Pixel intensity, shape, size, the velocity….)

↓

Segmentation an image into objects with different tools :

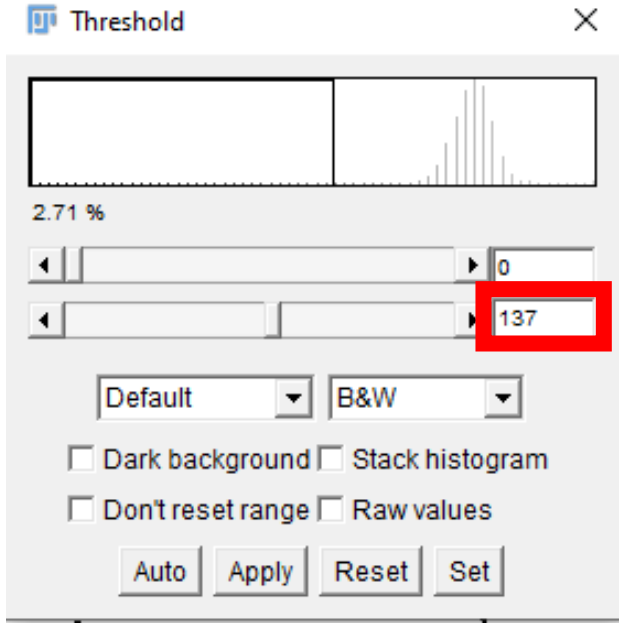
Threshold (pixel intensity)
Analyze particles (shape),
trackmate (shape, size and velocity)
…

2.2 Image processing with Fiji

2.2.3 Image treatment : segmentation based on pixel intensity

Fiji: image>Adjust> threshold

Thresholding is a technique for dividing an image into two (or more) classes of pixels, which are typically called “foreground” and “background.” Global thresholding works by choosing a value cutoff, such that every pixel less than that value is considered one class, while every pixel greater than that value is considered the other class.*



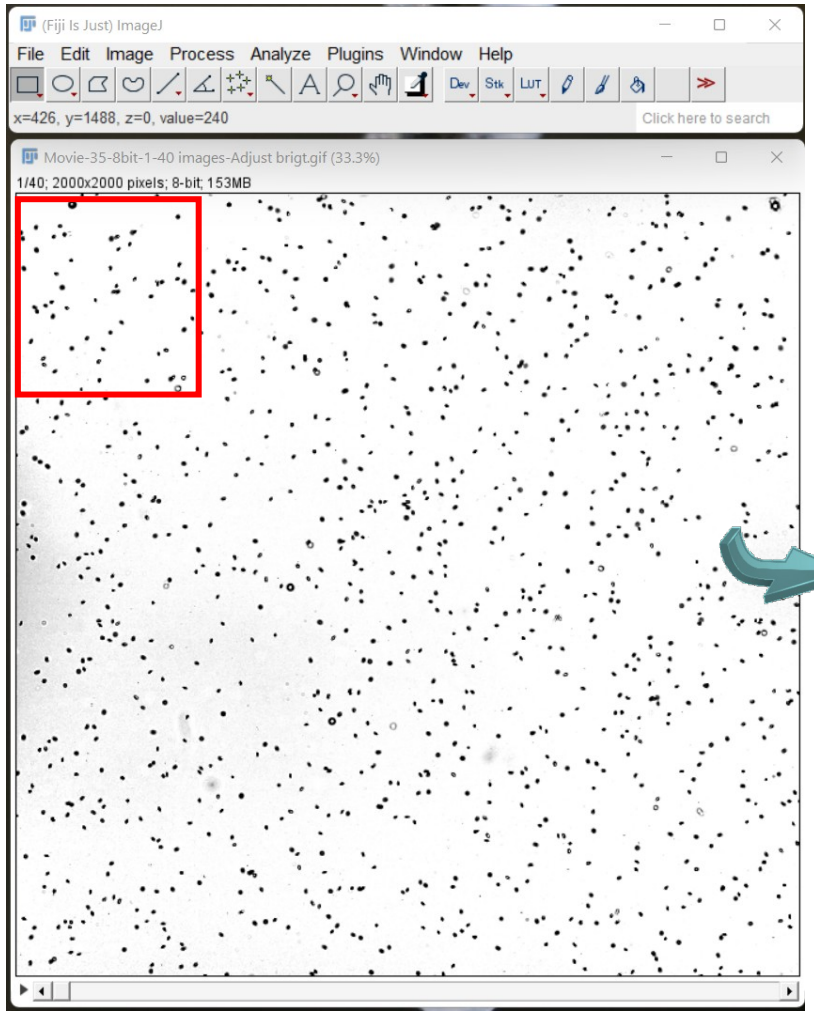
In our case we select “Auto” which automatically set the threshold at 137 (137 out of 255 pixel depth).
Below this value all pixel will be considere white, above it black.

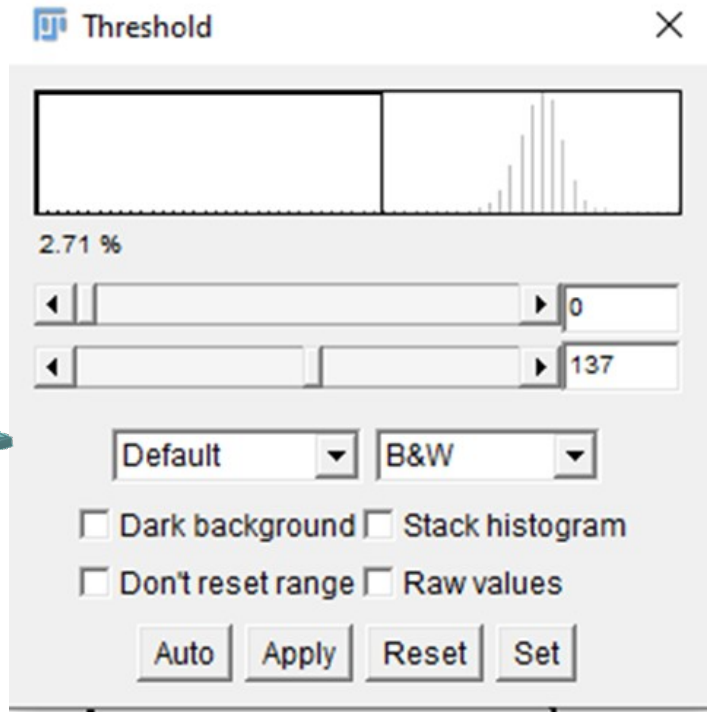
*https://imagej.net/imaging/thresholding

2.2 Image processing with Fiji

2.2.3 Image treatment : segmentation based on pixel intensity

Fiji: image>Adjust> threshold



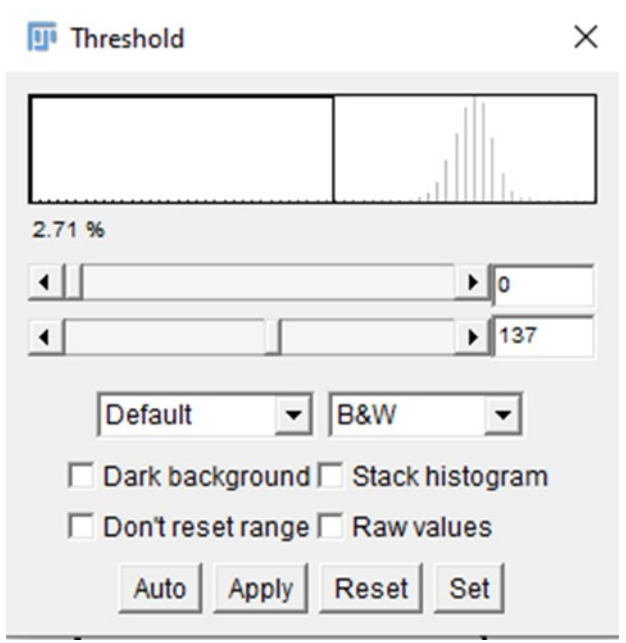


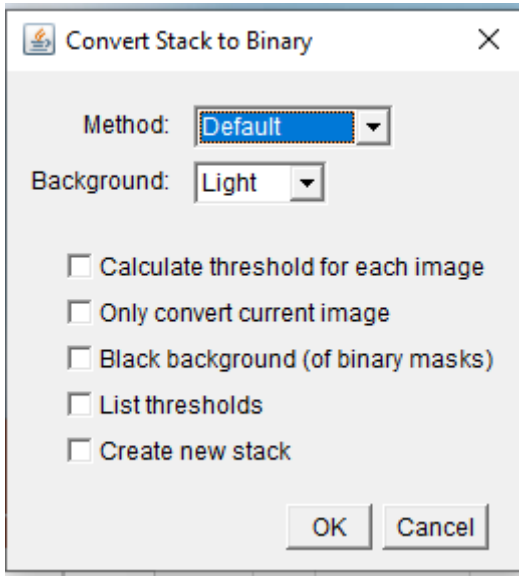
2.2 Image processing with Fiji

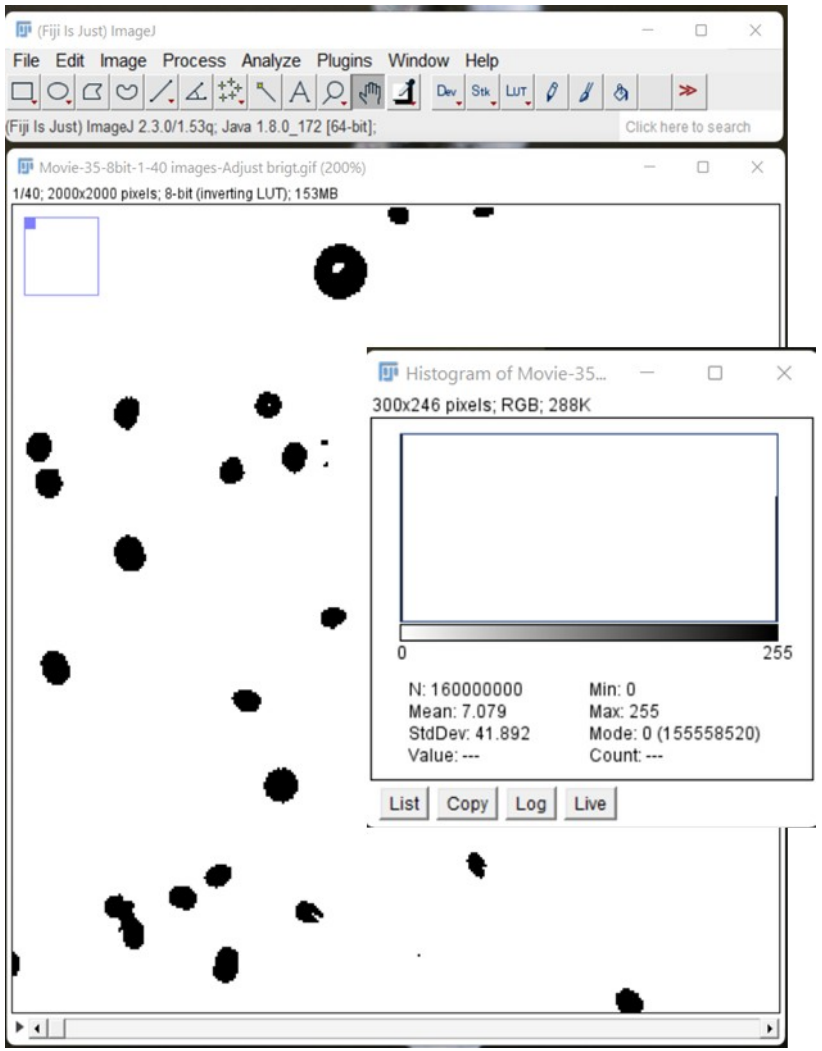
2.2.3 Image treatment : segmentation based on pixel intensity

Fiji: image>Adjust> threshold

a. Extract objects (cells) based on their pixel intensity







As result we extracted our zoospores cells that now appear as black objects (pixel intensity 255) over a white background (pixel intensity 0)

For more fun

- Tutorials for Fiji

  https://imagej.net/tutorials/
  https://www.youtube.com/watch?v=0C4HSKNsRYs
  https://www.youtube.com/watch?v=y567dfZ_reI
  https://www.youtube.com/watch?v=uuz7ivTnW9o&t=544s